

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

RIMFROST AS,
Petitioner,

v.

AKER BIOMARINE ANTARCTIC AS,
Patent Owner.

IPR2020-01532
Patent 9,644,169 B2

Before ERICA A. FRANKLIN, JON B. TORNQUIST, and
MICHAEL A. VALEK, *Administrative Patent Judges*.

TORNQUIST, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. *Background and Summary*

Rimfrost AS (“Petitioner”) filed a Petition (Paper 2, “Pet.”) requesting an *inter partes* review of claims 1–20 of U.S. Patent No. 9,644,169 B2 (Ex. 1001, “the ’169 patent”). Aker Biomarine Antarctic AS (“Patent Owner”) did not file a Preliminary Response to the Petition. Upon review of Petitioner’s arguments and evidence, we instituted an *inter partes* review of all claims and grounds asserted in the Petition (Paper 6, “Institution Decision” or “Inst. Dec.”).

Patent Owner subsequently filed a Response (Paper 9, “PO Resp.”), to which Petitioner filed a Reply (Paper 15, “Pet. Reply”), and Patent Owner filed a Sur-Reply (Paper 18, “Sur-Reply”). With authorization, Patent Owner also filed a paper identifying arguments and evidence in Petitioner’s Reply that it considers to be improper (Paper 19), to which Petitioner filed a response (Paper 21).

Petitioner also filed a motion to exclude certain evidence relied upon by Patent Owner (Paper 25), to which Patent Owner filed an opposition (Paper 26), and Petitioner filed a reply (Paper 28).

In support of their respective positions, Petitioner relies upon the declaration and reply declaration of Dr. Stephen J. Tallon (Exs. 1006, 1086), and Patent Owner relies upon the declaration of Dr. Jacek Jaczynski (Ex. 2015).

An oral hearing was held on January 12, 2022, and a transcript of the hearing is included in the record (Paper 32, “Tr.”).

B. Real Parties-in-Interest

Petitioner identifies itself, Olympic Holding AS, Emerald Fisheries AS, Rimfrost USA, LLC, Rimfrost New Zealand Limited, and Bioriginal Food and Science Corp. as real parties-in-interest. Pet. 3. Based on various ownership interests, and out of “an abundance of caution,” Petitioner also identifies Stig Remøy, SRR Invest AS, Rimfrost Holdings AS, and Omega Protein Corporation as real parties-in-interest. *Id.*

Patent Owner identifies itself as the real party-in-interest in this proceeding. Paper 5, 1.

C. Related Matters

The parties identify as a related matter *Aker Biomarine Antarctic AS v. Olympic Holding AS*, 1:16-CV-00035-LPS-CJB (D. Del.), which involved U.S. Patent Nos. 9,028,877 B2 (“the ’877 patent”) and 9,078,905 B2 (“the ’905 patent”). Pet. 3–4; Paper 5, 1. The parties further identify Investigation No. 337-TA-1019 by the United States International Trade Commission, which involved the ’877 and ’905 patents, as well as U.S. Patent No. 9,320,765 (“the ’765 patent”), U.S. Patent No. 9,375,453 (“the ’453 patent”), and U.S. Patent No. 9,072,752 (“the ’752 patent”). Pet. 4; Paper 5, 1–2.

The parties also identify the following Board proceedings as related matters:

- IPR2017-00745 and IPR2017-00747, which requested review of the ’905 patent (all challenged claims found unpatentable (Ex. 1103), decision affirmed on appeal (Ex. 1154));
- IPR2017-00746 and IPR2017-00748, which requested review of the ’877 patent (all challenged claims found unpatentable (Ex. 1104), decision affirmed on appeal (Ex. 1154));

- IPR2018-00295, which requested review of the '765 patent (all challenged claims found unpatentable (Ex. 1129));
- PGR2018-00033, which requested review of U.S. Patent No. 9,644,170 (institution denied because the challenged patent was not eligible for post grant review);
- IPR2018-01178 and IPR2018-01179, which requested review of the '453 patent (all challenged claims found unpatentable (Exs. 1157, 1158));
- IPR2018-01730, which requested review of the '752 patent (all challenged claims found unpatentable (Ex. 1159)); and
- IPR2020-01533, which requested review of U.S. Patent No. 9,816,046 B2 (pending).

Pet. 4–7; Paper 5, 2–4.

D. The '169 Patent

The '169 patent discloses extracts from Antarctic krill that include bioactive fatty acids. Ex. 1001, 1:22–23. The '169 patent explains that krill oil compositions, including compositions having up to 60% w/w phospholipid content and as much as 35% w/w EPA/DHA¹ content, were known in the art. *Id.* at 1:57–60. The '169 patent further explains that “[k]rill oil compositions have been described as being effective for decreasing cholesterol, inhibiting platelet adhesion, inhibiting artery plaque formation, preventing hypertension, controlling arthritis symptoms, preventing skin cancer, enhancing transdermal transport, reducing the

¹ According to the '169 patent, “EPA” is 5,8,11,14,17-eicosapentaenoic acid and “DHA” is 4,7,10,13,16,19-docosahexanoic acid. Ex. 1001, 9:12–16.

symptoms of premenstrual symptoms or controlling blood glucose levels in a patient.” *Id.* at 1:49–55.

According to the ’169 patent, frozen krill are typically transported from the Southern Ocean to a processing site, but lipases and phospholipases within the krill can result in the decomposition of glycerides and phospholipids during transport. *Id.* at 2:6–16, 9:61–10:8. To avoid the problem of enzymatic decomposition of krill products, the ’169 patent describes a method of thermally denaturing the lipases and phospholipases in fresh-caught krill prior to storage and processing. *Id.* at 10:5–10, 10:45–50. The ’169 patent reports that these denaturing steps allow for the storage of krill material “for from about 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, or 12 months to about 24 to 36 months prior to processing.” *Id.* at 10:36–44.

After denaturation, the krill can be subject to extraction processes either on board the ship or at a remote location. *Id.* at 10:38–40. In one embodiment, krill oil is extracted from krill meal in two stages. *Id.* at 9:54–55. In the first stage, a neutral fraction is extracted using either neat supercritical CO₂ or CO₂ in combination with 5% ethanol. *Id.* at 9:55–57. In the second stage, polar lipids (phospholipids) are extracted by adding at least 20% ethanol to the supercritical CO₂ extraction medium. *Id.* at 9:58–60. The ’169 patent reports that “[k]rill oil extracted from denatured krill meal by supercritical fluid extraction even 19 months after the production of the meal contained virtually no decomposed phospholipids.” *Id.* at 11:3–6. The ’169 patent further reports that the novel krill oil compositions of the invention are “characterized by containing high levels of astaxanthin, phospholipids, includ[ing] enriched quantities of ether phospholipids, and omega-3 fatty acids.” *Id.* at 9:46–49.

E. Illustrative Claims

Petitioner challenges claims 1–20 of the '169 patent. Claims 1 and 12 are independent, with claims 2–11 depending, directly or indirectly from claim 1, and claims 13–20 depending, directly or indirectly, from claim 12. Claims 1 and 4 are illustrative of the challenged claims and are reproduced below:

1. A method of production of krill oil comprising:
 - a) providing krill;
 - b) treating said krill to denature lipases and phospholipases in said krill to provide a denatured krill product;
 - c) storing said denatured krill product for a storage period of from 1 to 24 months;
 - d) after said storage period, extracting oil from said denatured krill product with a polar solvent to provide a krill oil with from about 3% to about 15% ether phospholipids w/w of said krill oil [and] astaxanthin esters in amount of greater than about 100 mg/kg of said krill oil.

Ex. 1001, 35:45–55.

4. The method of claim 1, wherein said denatured krill product is a krill meal.

Id. at 36:12–13.

F. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–20 would have been unpatentable on the following grounds:

Claims Challenged	35 U.S.C. §	References/Basis
1–5, 7–15, 17–20	103(a) ²	Breivik II, ³ Catchpole, ⁴ Budziński, ⁵ Fricke, ⁶ Randolph ⁷
6, 16	103(a)	Breivik II, Catchpole, Budziński, Fricke, Randolph, Sampalis I ⁸

II. ANALYSIS

A. Legal Standards

A patent claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter as a whole would have been obvious at the time the

² The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) (“AIA”), amended 35 U.S.C. §§ 102 and 103. Because the claims of the ’169 patent have an effective filing date before the March 16, 2013, effective date of the applicable AIA amendments, we refer to the pre-AIA versions of 35 U.S.C. §§ 102 and 103.

³ WO 2008/060163 A1, published May 22, 2008 (Ex. 1037) (“Breivik II”). Breivik II claims priority to U.S. Provisional Application No. 60/859,289, filed November 16, 2006. Ex. 1037, code (30).

⁴ WO 2007/123424 A1, published Nov. 1, 2007 (Ex. 1009) (“Catchpole”).

⁵ Budziński et al., *Possibilities of processing and marketing of products made from Antarctic krill*, FAO Fish. Tech. Pap. (268):46 (1985) (Ex. 1008).

⁶ Fricke et al., *Lipid, Sterol and Fatty Acid Composition of Antarctic Krill (Euphausia superba Dana)*, 19(11) LIPIDS 821–827 (1984) (Ex. 1010) (“Budziński”).

⁷ US 2005/0058728 A1, published Mar. 17, 2005 (Ex. 1011) (“Randolph”).

⁸ Sampalis et al., *Evaluation of the Effects of Neptune Krill Oil™ on the Management of Premenstrual Syndrome and Dysmenorrhea*, 8(2) ALT. MED. REV. 171–179 (2003) (Ex. 1012) (“Sampalis I”).

invention was made to a person having ordinary skill in the art to which said subject matter pertains. *See KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) if in the record, objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

B. Level of Ordinary Skill in the Art

Petitioner provides the following definition of the ordinarily skilled artisan

A [person of ordinary skill in the art] would have held an advanced degree in marine sciences, biochemistry, organic (especially lipid) chemistry, chemical or process engineering, or associated sciences with complementary understanding, either through education or experience, of organic chemistry and in particular lipid chemistry, chemical or process engineering, marine biology, nutrition, or associated sciences; and knowledge of or experience in the field of extraction. In addition, a [person of ordinary skill in the art] would have had at least five years applied experience.

Pet. 9–10 (citing Ex. 1006 ¶¶ 35–36).

Neither Patent Owner nor Dr. Jaczynski dispute Petitioner’s definition of a person of ordinary skill in the art. Ex. 2015 ¶ 8 (“For purposes of this proceeding, I will accept [Petitioner’s] definition of a [person of ordinary skill in the art].”).

Upon review of the ’169 patent and the prior art of record, we find Petitioner’s uncontested definition reasonable and supported by the evidence of record. Accordingly, we adopt this definition of one of ordinary skill in the art for purposes of this Decision.

C. Claim Construction

Petitioner provides proposed claim constructions for the terms “krill oil,” “denature lipases and phospholipases,” “polar solvent,” “astaxanthin esters,” “krill meal,” and “freshly harvested.” Pet. 32–39. With respect to “krill meal,” Petitioner contends this term should be construed to mean “processed krill with reduced water content from which krill oil can be extracted.” *Id.* at 38.

Patent Owner provides proposed constructions for the terms “krill meal” and “denature lipases and phospholipases.” PO Resp. 7–12. With respect to “krill meal,” Patent Owner contends Petitioner’s proposed construction of “krill meal” is overbroad and ignores the ordinary and customary meaning of a meal, which is a “powder formed by particle size reduction of a starting material.” *Id.* at 8 (citing Ex. 2015 ¶¶ 14–16). Thus, Patent Owner construes “krill meal” to mean “a krill powder resulting from the processing of krill.” *Id.*

Upon review of the parties’ arguments and supporting evidence, we determine that no terms of the ’169 patent require construction in order to resolve the disputes in this proceeding. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999) (“[O]nly those terms need be construed that are in controversy, and only to the extent necessary to resolve the controversy.”)).

D. Collateral Estoppel

Petitioner contends that in three prior IPRs the Board analyzed similar subject matter to that claimed in the ’169 patent and found the claims unpatentable based on the same prior art asserted in this proceeding. Pet. 40–41. “Given the materially identical claim limitations and common

references,” Petitioner contends “the Board’s factual findings and conclusions of law regarding, *inter alia*, the teachings of the prior art, the motivation to combine those teachings, and claim construction in the prior Final Written Decisions should be given preclusive effect.” *Id.* at 41–42.

Petitioner concedes that the precise issue of whether one of ordinary skill in the art would have understood that a denatured krill product could be stored for 1 to 24 months before extraction was not at issue in the prior IPRs, but contends collateral estoppel still applies because the “inclusion of an intuitive and common sense storage limitation does not materially alter the question of the ’169 patent’s unpatentability.” *Id.* at 41; Pet. Reply 7.

“Collateral estoppel protects a party from having to litigate issues that have been fully and fairly tried in a previous action and adversely resolved against a party-opponent.” *Ohio Willow Wood Co. v. Alps South, LLC*, 735 F.3d 1333, 1342 (Fed. Cir. 2013). In the context of patent claims, collateral estoppel applies “[i]f the differences between the unadjudicated patent claims and adjudicated patent claims do not materially alter the question of invalidity.” *Id.*

Petitioner points to no prior decision of the Board where the issue of whether it would have been obvious to denature a krill product and then store it for 1 to 24 months before extraction was at issue. Moreover, to demonstrate that the added limitation would have been obvious, Petitioner relies heavily on the disclosures of Budziński, which was not at issue or analyzed in the prior proceedings. Pet. 45; PO Resp. 13 (asserting that the prior art in the previous proceedings “did not include Budziński”). Accordingly, Petitioner has not demonstrated that collateral estoppel applies in this proceeding.

E. Reliance on Breivik II

Patent Owner contends the invention recited in the claims of the '169 patent was conceived and reduced to practice prior to the earliest possible priority date of Breivik II. PO Resp. 17. And, because each ground asserted in the Petition relies on Breivik II, Patent Owner contends the Petition must be denied. *Id.* at 36.

Petitioner asserts that Patent Owner has failed to antedate Breivik II and, even if Breivik II is antedated, Petitioner has shown that the challenged claims are unpatentable without that reference, as “one or more alternative references are provided for every limitation for which Breivik II is cited.” Pet. Reply 7.

We begin by considering whether Petitioner’s challenges based on Breivik II, Catchpole, Budziński, Fricke, and Randolph necessarily depend upon the teachings and disclosures of Breivik II. Based upon our review of the Petition, we find that they do not because Petitioner consistently relies on multiple prior art references to establish that a particular limitation is disclosed in the prior art, and at no point does it rely solely on Breivik II as describing any particular limitation. Pet. 44–49. For example, in addressing whether it was known to treat krill to provide a denatured product, Petitioner asserts that “[t]reating krill to provide a denatured krill product was well known, disclosed and taught in Breivik II, Budziński, and Fricke.” *Id.* at 44.

Based upon our review of the Petition, we also find that there is not any particular reason to combine the prior art references set forth by Petitioner that relies solely on Breivik II. *See id.* at 59–66. For example, in asserting that one of ordinary skill in the art would have understood that krill oil could be extracted using methods known in the art, Petitioner relies on

the independent disclosures of Brevik II, Catchpole, Budziński, and Fricke. *Id.* at 62–63.

Given the structure of the Petition and its reliance on multiple, independent references to establish obviousness, we find that the Petition does not necessarily fail if Breivik II is removed from consideration. Moreover, because (for the reasons set forth below) we determine that Petitioner has shown by a preponderance of the evidence that the challenged claims would have been obvious over Catchpole, Budziński, Fricke, and Randolph, without the teachings of Breivik II, we do not rely on Breivik II for our analysis. Thus, the issue of whether Patent Owner’s evidence of conception and reduction to practice is sufficient to antedate Breivik II is moot, and we do not resolve the issue in our Decision.

F. Claims 1–5, 7–15, and 17–20 over Catchpole, Budziński, Fricke, and Randolph

Petitioner contends that the subject matter of claims 1–5, 7–15, and 17–20 would have been obvious over the combined disclosures of Catchpole, Budziński, Fricke, and Randolph. Pet. 42–66.

1. Catchpole

Catchpole discloses “a process for separating lipid materials containing phospholipids.” Ex. 1009, 1:5–6, 3:27–28. Catchpole explains that phospholipids “have been implicated in conferring a number of health benefits including brain health, skin health, eczema treatment, anti-infection, wound healing, gut microbiota modifications, anticancer activity, alleviation of arthritis, improvement of cardiovascular health, and treatment of metabolic syndromes.” *Id.* at 1:29–2:2. Catchpole further explains that products having high levels of particular phospholipids “may be employed

in a number of applications, including infant formulas, brain health, sports nutrition and dermatological compositions.” *Id.* at 25:9–13.

In Example 18, Catchpole describes the fractionation of krill lipids from freeze-dried krill powder using a process that employs supercritical CO₂ in a first extraction and then CO₂ and ethanol in a second extraction. *Id.* at 24:1– 16. Table 16 of Catchpole is reproduced below.

Table 16

	Yield % of feed	Composition, %							Other compounds
		PC	PI	PS	PE	CL	AAPC	AAPE	
Feed		6.6	0.0	0.0	0.4	0.1	0.6	0.1	78.6
Extract 2	4.3	39.8	0.0	0.0	0.3	0.2	4.6	0.2	53.7
Residue	79.2	3.6	0.0	0.0	0.3	0.2	0.5	0.1	93.4

Table 16 reports the phospholipid concentrations present in the krill oil extract obtained in Example 18. *Id.* at Table 16. As shown in Table 16, Extract 2 has a total phospholipid concentration of 45.1%, and includes 39.8% phosphatidylcholine (“PC”), 4.6% alkylacylphosphatidylcholine (“AAPC”), and 0.2% alkylacylphosphatidylethanolamine (“AAPE”). *Id.*; *see also id.* at 1:11–15 (defining PC, PI, PS, PE, and CL).

2. Budziński

Budziński, published in 1985, “presents the current state of possible commercial-scale uses of krill-processing technologies,” including processing technologies for products for “human consumption, animal feed, and industrial purposes.” Ex. 1008, Abstr. Budziński explains that the “main difficulty in krill processing” is caused by proteolytic enzymes, such as lipases. *Id.* at 6.⁹ According to Budziński, lipases “cause the

⁹ Unless otherwise noted, citations to prior art references are to the original page numbers of the documents, not to the page numbers added by Petitioner in the lower right corner of the documents.

decomposition of phospholipids and, to a lesser degree, triglycerides,” and result in an increase in free fatty acids in krill oil. *Id.* at 6–7. To avoid enzyme degradation of krill products, Budziński explains that storage time for raw krill products “should be as short as possible” and that vessels “must be properly equipped for processing” krill as soon as possible after capture. *Id.* at 9, 20 (“Efforts should be made to regulate catches so that the time before processing is as short as possible.”), 25 (“Due to its technological properties, the raw materials should be processed as soon as possible after capture. The only way to meet this requirement is to install processing facilities on-board the vessel.”). Budziński discloses that the optimum temperature for heating krill is believed to be 80–85 °C and that properly processed krill meal is “stable for 13 months of storage without addition of antioxidants.” *Id.* at 20–22.

3. *Fricke*

Fricke reports the total lipid content and composition for krill samples caught in the Antarctic Ocean in December 1977 and March 1981.

Ex. 1010, Abstr., 822. Fricke reports that “[k]rill samples of 5 kg were quick-frozen and stored at -35 C until analyzed” and that “lipid extraction was performed according to Folch et al.” *Id.* at 821. The samples analyzed in Fricke contained approximately 33.3% +/- 0.5% w/w (1977 sample) and 40.4% +/- 0.1% w/w (1981 sample) triacylglycerols and 16.1 (1977 sample) and 8.5 (1981 sample) percent free fatty acids. *Id.* at 822.

Fricke notes that samples from the 1977 sample contained free fatty acid levels that were about twice that of the 1981 sample. *Id.* at 822. In contrast, samples of the same haul that had been cooked on board “immediately after hauling,” showed a free fatty acid content that “was much lower, ranging from 1% to 3% of total lipids.” *Id.* at 822–23.

4. *Randolph*

Randolph discloses compositions for modulating an inflammatory or immunomodulatory response, including krill oil. Ex. 1011 ¶ 8. The krill oil of Randolph may be obtained from any member of the *Euphausia* family, including *Euphausia superba*, and typically contains “between about 300 mg and about 3000 mg of a krill oil ingredient.” *Id.* ¶¶ 39–40. Randolph further teaches that the composition can contain any amount of an astaxanthin ingredient, including at least about 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 35, 40, 50, 60, 70, 80, or 90 percent astaxanthin, typically resulting in between about 0.5 mg and about 50 mg of astaxanthin in the product. *Id.* ¶ 44.

Randolph explains that the ingredients of the composition can be processed into various “delivery systems,” such as capsules, soft gel capsules, tablets, gel tabs, lozenges, strips, granules, powders, concentrates, solutions, lotions, creams or suspensions. *Id.* ¶¶ 46, 52.

5. *Analysis of Claim 1*

a) “providing krill”

Petitioner contends Fricke and Budziński teach or suggest providing krill. Ex. 1010, 822 (Fricke expressly disclosing catching krill (*Euphausia superba*)); Ex. 1008, 2 (Budziński disclosing providing “freshly harvested” krill); Ex. 1006 ¶¶ 439–440.

Patent Owner does not contest that providing krill was known in the art.

Upon review of the prior art of record, we find that “providing krill” was well known in the art and is expressly disclosed in Fricke and Budziński. Ex. 1010, 822; Ex. 1008, 2; *see also* Ex. 1032, 18 (Grantham disclosing the catching of krill); Ex. 1033 ¶ 5 (Yoshitomi disclosing the catching krill).

b) *“treating said krill to denature lipases and phospholipases in said krill to provide a denatured krill product”*

Petitioner contends treating krill to denature lipases and phospholipases to form a denatured krill product was well known in the art, as evidenced by both Budziński and Fricke. Pet. 44. We agree.

Budziński and Fricke both disclose the problem of decomposition of useful krill products due to krill’s very active system of proteolytic enzymes. For example, Budziński discloses that “[t]he main difficulty in krill processing is caused by the very active system of proteolytic enzymes” and lipases in krill, which cause the decomposition of phospholipids and, to a lesser extent, triglycerides. Ex. 1008, 6. Budziński further discloses that the result of enzyme activity “is an increase of free fatty acids (FFA)” within krill products. *Id.* Fricke similarly discloses that “residual lipolytic activity against phospholipids exists even at temperatures of -30 C and below” and result in increased levels of free fatty acids. Ex. 1010, 822.

Both Budziński and Fricke also teach that cooking krill can denature the problematic lipases and phospholipases within krill, with Budziński further disclosing that the optimal cooking temperature is believed to be 80–85 °C. Ex. 1008, 20; Ex. 1010, 822–823; Ex. 1006 ¶¶ 439–440 (Dr. Tallon testifying that the cooking procedures disclosed in Fricke and Budziński would result in a denatured krill product).

Patent Owner does not contest that it was well known in the art that krill enzymes needed to be denatured soon after hauling krill on board a ship and that heating, as disclosed in Budziński and Fricke, was a standard process for denaturing krill enzymes that would result in a denatured krill product.

Upon review of the parties' arguments and evidence, we find that one of ordinary skill in the art would have understood from the disclosures of Budziński and Fricke that krill enzymes should be denatured soon after krill are caught and that this process would form a denatured krill product. Ex. 1008, 9 (Budziński noting that processing of krill should occur as soon as possible after a catch), 20; Ex. 1010, 822–823.

c) *“storing said denatured krill product for a storage period of from 1 to 24 months”*

Petitioner contends that processing methods involving storage of a denatured krill product for from 1 to 24 months are taught or suggested in both Budziński and Fricke. Pet. 45. We agree.

Section 4.5 of Budziński discusses the standard method for producing krill meal, as well as the properties of the resulting krill meal. Ex. 1008, 18–22. In subsection 4.5.1, Budziński specifically notes that to increase yields, krill should not be stored for longer than 8–10 hours before processing, that the traditional method for krill meal production was “cooking and pressing or centrifuging and drying” the krill, and that an optimal cooking temperature of 80–85 °C should be used to denature the krill. *Id.* at 20.

In Section 4.5.2, Budziński notes that denatured krill meal has a red-pink color caused by the presence of astaxanthin and its esters, and that denatured krill meal had 55 µg/kg astaxanthin after seven months of storage. *Id.* at 21. Budziński further notes that, despite concerns that the highly unsaturated lipids in krill meal could be susceptible to oxidation, “[e]xtensive investigations” had shown that krill meal is “stable for 13 months of storage without addition of antioxidants.” *Id.* at 21–22.

Fricke discloses quick-freezing cooked krill and then storing these samples before analysis. Ex. 1010, 821–823. Fricke further discloses that

samples that were cooked before freezing had a much lower level of free fatty acids than krill frozen immediately after hauling onboard. *Id.* at 822–823.

Petitioner, supported by the testimony of Dr. Tallon, asserts that given the disclosures of Budziński and Fricke, one of ordinary skill in the art would have understood that a denatured krill product could be stored for up to 13 months before extraction. Pet. 45, 65; Ex. 1006 ¶¶ 441–444.

Patent Owner argues that Fricke contains no disclosure of how long the denatured krill were stored after being frozen, but does not contest Petitioner’s assertion that Budziński expressly teaches denatured krill meals that have been stored for at least 13 months.¹⁰ PO Resp. 38.

On this record, we agree with Petitioner that one of ordinary skill in the art, considering the disclosures of Fricke and Budziński, would have understood that a denatured krill product could be successfully stored for up to 13 months.

d) *“after said storage period, extracting oil from said denatured krill product with a polar solvent”*

Petitioner contends extracting krill oil with a polar solvent was well known in the art, as expressly described in Catchpole, Budziński, and Fricke, and that one of ordinary skill in the art would have understood that a denatured krill product stored for up to 13 months could be successfully extracted using a polar solvent. Pet. 46. We agree.

¹⁰ As discussed below, Patent Owner contests whether Budziński, or any other individual prior art reference, teaches or suggests denaturing a krill product, storing that product for 1 to 24 months, and then extracting krill oil from the denatured krill product. PO Resp. 36–40. Patent Owner also questions the quality of the krill meal stored for 13 months in Budziński. *Id.* at 38–39. We address those arguments in Section II.F.5.(d) below.

Fricke expressly discloses cooking a krill product to denature krill enzymes, storing this denatured product for a period of time, and then extracting a lipid fraction from the denatured krill product using a polar solvent. Ex. 1010, 821–823. Although the specific storage period is not expressly recited in Fricke, given the location of the vessel during krill capture and the fact that the krill were frozen into blocks, we agree with Dr. Tallon that one of ordinary skill in the art would have understood that the denatured krill was stored for more than a trivial period of time prior to extraction. Ex. 1006 ¶¶ 310–313; Ex. 1086 ¶ 61.

Budziński discloses a krill meal that is stable for at least 13 months and that krill oil can be extracted from krill products using various organic solvents. Ex. 1008, 21–22, 24. Although Budziński does not expressly disclose extracting krill oil from a denatured krill product, such as krill meal that has been stored for 1 to 24 months, Dr. Tallon explains that because the denatured krill meal of Budziński is stable for at least 13 months, one of ordinary skill in the art would have understood that this krill meal could “be extracted just the same as a fresh krill meal” and would provide a usable krill oil product. Ex. 1006 ¶¶ 168–171.

Catchpole discloses extracting krill oil from a freeze-dried krill powder using CO₂ and ethanol, a polar solvent. Ex. 1009, 24:1–16.

In view of the disclosures of Fricke, Budziński, and Catchpole, Petitioner contends one of ordinary skill in the art would have understood that denatured krill products could be stored for up to 13 months before successfully extracting a krill oil using a polar solvent. Pet. 46, 61.

Patent Owner argues that none of the identified prior art references discloses using a polar solvent to extract krill oil from a denatured krill product that has been stored from 1 to 24 months. PO Resp. 37–39.

According to Patent Owner, Catchpole provides no storage information for its freeze-dried krill powder, Fricke “contains no teaching as to the storage time for the cooked krill,” and Budziński does not teach solvent extraction from a meal after any period of storage or that its krill meal was “stable” with respect to krill oil extraction, as opposed to use as an animal feed. *Id.* at 38–39.

When considering the question of obviousness, we consider what the disclosures of the prior art, as a whole, would have taught or suggested to one of ordinary skill in the art. *See Bradium Techs. v. Iancu*, 923 F.3d 1032, 1050 (Fed. Cir. 2019). We do not limit our analysis to what each reference discloses in isolation. *Id.* (“A finding of obviousness, however, cannot be overcome ‘by attacking references individually where the rejection is based upon the teachings of a combination of references.’”). Here, Catchpole expressly discloses extracting krill oil from a krill powder (which is a “krill meal” under either parties’ understanding of that term) using a polar solvent. Ex. 1009, 24:1–16. Fricke discloses denaturing krill, storing this krill for a period of time¹¹, and then extracting a lipid fraction from the denatured krill using a polar solvent. Ex. 1010, 821–823. And Budziński discloses both

¹¹ Fricke does not disclose the precise storage period for its denatured krill. Ex. 1010, 822–823. A subsequent paper by Fricke suggests, however, that this product was stored for “some months” prior to extraction. Ex. 1160, 131. Patent Owner disputes whether Fricke’s subsequent disclosure that the denatured krill sample was stored for “some months” may be relied upon in this Decision, as it was not cited in the Petition. Paper 19, 2. We need not address this issue because we agree with Dr. Tallon that it is evident from Fricke itself that the denatured krill samples were stored for some time before extraction, and Budziński expressly discloses a krill meal that is stable for at least 13 months of storage. Ex. 1006 ¶¶ 167, 311. As such, our finding as to this claim element would be the same whether Fricke’s subsequent disclosures (Ex. 1160) were considered or not.

that polar solvents may be used to extract krill oil and that denatured krill meal is stable for at least 13 months of storage. Ex. 1008, 21–22, 24.

Petitioner persuasively explains why one of ordinary skill in the art considering these disclosures would have understood that krill can be denatured to form a krill product, stored for up to 13 months, and then extracted using a polar solvent, as recited in claim 1 of the '169 patent. Pet. 44–46, 60–63; Ex. 1006 ¶¶ 168–171, 306–313, 442–449.

As noted above, Patent Owner argues that the krill meal of Budziński was stable only with respect to use as an animal feed, which has different stability requirements than a krill meal that is used to produce a krill oil. PO Resp. 46–47. Dr. Tallon persuasively explains, however, that the presence of astaxanthin in the krill meal of Budziński demonstrates that the krill meal was stable with respect to both enzymatic breakdown and oxidation. Ex. 1006 ¶ 167; Ex. 1086 ¶ 47 (Dr. Tallon testifying that the presence of astaxanthin demonstrated that the krill meal was stable with respect to oxidation and enzyme activity). Dr. Tallon also persuasively explains that given the presence of astaxanthin in the krill meal, as well as Budziński's report that the krill meal was "stable," one of ordinary skill in the art would have understood that the stored krill meal of Budziński was suitable for krill oil extraction. Ex. 1006 ¶¶ 167–168, 171 (Dr. Tallon explaining that the process of extraction of krill meal is the same regardless of whether it is fresh or stored for an extended period of time); Ex. 1086 ¶ 47. Thus, we find that one of ordinary skill in the art would have understood that Budziński's krill meal was stable with respect to both oxidation and enzymatic breakdown and was suitable for extraction to produce krill oil.

Patent Owner also argues that Budziński teaches that “since extraction costs are high and special installations are needed, it is unrealistic to expect production of krill oil in practice.” PO Resp. 38–39 (quoting Ex. 1008, 24). According to Patent Owner, if anything, this disclosure “teaches away from extracting oil from krill.” *Id.* at 39. Although Budziński does assert that krill oil production is unrealistic due to the high extraction costs, it also notes the high potential value of krill oil products, and subsequent references disclose successfully extracting krill oil for use in pharmaceuticals or nutritional products. Ex. 1008, 24 (Budziński noting that “krill oil could be a valuable raw material for oil, cosmetic and pharmaceutical industries”); Ex. 1009, 24:1–17; Ex. 1011 ¶¶ 39–40, 44; Ex. 1012, 174 (“Neptune Krill Oil TM (NKO TM) is a natural health product extracted from Antarctic krill”); Ex. 1075, 5–6. As such, we agree with Petitioner that due to subsequent advances in the art that occurred prior to the earliest possible priority date of the ’169 patent, Budziński would not have taught away from, or counseled against, producing krill oil from a denatured krill product. Pet. Reply 21–22.

In view of the foregoing, we find that one of ordinary skill in the art would have understood, based on the disclosures of Budziński, Fricke, and Catchpole, that a denatured krill meal could be stored for up to 13 months before using a polar solvent to extract a useful krill oil.

e) *“to provide a krill oil with from about 3% to about 15% ether phospholipids w/w of said krill oil”*

Petitioner contends that ether phospholipids are naturally present in krill and could be extracted using “conventional extraction techniques and known solvent systems,” such as those used in Catchpole. Pet. 47 (citing Ex. 1006 ¶ 450). Petitioner further contends that Catchpole expressly

describes the fractionation of krill lipids to form an extract containing “at least 4.8% ether phospholipids.” *Id.* (citing Ex. 1009, 14:7–11; Ex. 1006 ¶¶ 190–196).

Patent Owner does not contest Petitioner’s arguments with respect to this claimed subject matter. PO Sur-Reply 25. Indeed, inventor Dr. Snorre Tilseth testifies in support of Patent Owner’s argument regarding the prior conception of this limitation that “ether phospholipids are natural components of the krill oil which can be extracted in desired amounts by known methods.” Ex. 2001, 13.

In view of the foregoing, we find that it was known in the art to produce a krill oil having from about 3% to about 15% ether phospholipids w/w and that one of ordinary skill in the art would have understood that this krill oil could be obtained from a stable krill meal using conventional extraction techniques, including the use of a polar solvent.

f) “astaxanthin esters in an amount of greater than 100 mg/kg of said krill oil”

Petitioner contends that Randolph discloses krill oil compositions having at least 167 mg/kg of astaxanthin, which is equivalent to 158 mg/kg astaxanthin esters. Pet. 48.

Patent Owner does not contest that astaxanthin esters in amounts greater than 100 mg/kg of said krill oil were known in the art.

Upon review of the record as a whole, we find that one of ordinary skill in the art would have understood that krill oil with astaxanthin esters in an amount greater than 100 mg/kg were known in the art and disclosed in Randolph.

g) Reason to Combine

Petitioner contends Catchpole, Budziński, Fricke, and Randolph each relate to the same field of endeavor and that one of ordinary skill in the art would have sought to combine the disclosures of these references to produce the krill oil recited in claim 1 of the '169 patent in view of the known health benefits of krill products. Pet. 59.

In particular, Petitioner contends that one of ordinary skill in the art would have understood from the disclosures of Budziński and Fricke that the lipases and phospholipases in fresh krill would need to be promptly denatured or destroyed to prevent the formation of undesirable free fatty acids and degraded phospholipids. *Id.* at 59–60. Once the krill products are denatured, Petitioner contends one of ordinary skill in the art would have understood from the disclosures of Budziński and Fricke that the denatured krill products could be stored for at least 13 months, and would have sought to do so in order to extract the krill oil on land as opposed to at sea.

Id. at 60–61. According to Petitioner, one of ordinary skill in the art would have understood that land-based extraction would be beneficial because it would avoid the “hazards associated with using flammable polar solvents at sea” and “the expense of installing and operating extraction processes on a ship.” *Id.* at 61. Supporting Petitioner’s arguments, Dr. Tallon testifies that land-based extraction of krill oil would also provide “for cheaper installation and access to fuel needed to run the extraction process” and allow one to “extract at ones future time of choice to manage inventory and cash flow etc.” Ex. 1006 ¶ 74.

Once on land, Petitioner contends one of ordinary skill would have understood from the disclosures of Catchpole that supercritical fluid extraction with CO₂ and a polar solvent is a useful method of extracting krill

oil from krill products. Pet. 62. Petitioner further contends that it was understood in the art that ether phospholipids and astaxanthin had health benefits, could be extracted from denatured and previously-stored krill products, and that the oil extracted from denatured krill would be expected to have from about 3% to about 15% ether phospholipids w/w (Catchpole) and greater than about 100 mg/kg astaxanthin esters (Randolph). *Id.* at 63–66.

Patent Owner contends Petitioner’s rationale for storing denatured krill products for between 1 and 24 months, e.g., to avoid the dangers of using volatile solvents on a ship, is expressly contradicted by Breivik II¹² and the deposition testimony of Dr. Tallon. PO Resp. 40. In particular, Patent Owner contends that Breivik II expressly teaches that solvent-based extraction systems may be used on krill fishing vessels and that immediate processing and extraction after krill are hauled on board avoids hydrolysis and oxidation. *Id.* at 41. Patent Owner further asserts that Dr. Tallon admitted that the risks involved with using solvents on a fishing vessel could be successfully managed. *Id.* at 42–43.

As noted by Patent Owner, the evidence of record suggests that one of ordinary skill in the art would have known that krill processing—both denaturation and krill oil extraction—could be successfully performed on board a ship. Ex. 1037, 3–4; Ex. 2019, 49:18–25 (Dr. Tallon testifying

¹² Because Patent Owner relies on various disclosures of Breivik II as potentially counselling against combining the references to arrive at the claimed subject matter, we will consider these teachings in our Decision. We do not, however, consider Breivik II or Petitioner’s arguments with respect to this reference in determining whether the prior art teaches or suggests every limitation of the challenged claims.

during cross-examination that flammable solvents can be successfully managed on a ship). At the same time, Petitioner and Dr. Tallon explain that the dangers of using flammable solvents on a vessel are still a relevant consideration. Ex. 1086 ¶¶ 97–99; Ex. 2019, 49:18–25 (Tallon Deposition). Moreover, Dr. Tallon also persuasively identifies several other benefits of land-based krill oil extraction, such as the reduced costs of outfitting each ship with its own processing equipment (which is often essentially identical to that used on land), the ability to extract krill oil to meet specific demand in the market, and cheaper access to fuel. Pet. 61 (citing Ex. 1006 ¶¶ 173, 443); *see KSR*, 550 U.S. at 418 (noting that market forces and design needs can provide the reason to combine or modify prior art references). Patent Owner presents no argument or evidence to refute these additional benefits of land-based extractions, other than to note that Dr. Tallon does not cite any specific documents to support his assertions. PO Resp. 45 (Patent Owner noting the testimony of Dr. Tallon). Upon review of the evidence of record, we credit the testimony of Dr. Tallon that one of ordinary skill in the art would have understood that, in addition to avoiding the need to manage the use of flammable solvents at sea, there were a number of other benefits to land-based extractions of krill oil that would have motivated a skilled artisan to store the denatured krill products for between 1 and 24 months, as recited in claim 1.

Patent Owner also argues that Budziński “specifically teaches that the ‘raw material should be processed as soon as possible after capture’” and that the “only way to meet this requirement is to install processing facilities on-board the vessel.” PO Resp. 46 (quoting Ex. 1008, 31). According to Patent Owner, this is consistent with Breivik II’s “teaching of a process suitable for extraction of fresh krill on board the fishing vessel.” *Id.* (citing

Ex. 2015 ¶¶ 60–61). Patent Owner further argues that there are ample reasons to extract krill oil from fresh krill immediately after hauling them on board, as opposed to extracting oil from stored krill meal, such as the ability to avoid oxidation and polymerization materials in the krill oil. *Id.* at 45 (citing Ex. 1037, 2, 6, 10, 12).

As noted by Patent Owner, Budziński discloses that krill should be processed immediately after being hauled on board due to the “very active system of proteolytic enzymes” present in fresh caught krill. Ex. 1008, 6, 25. This processing, however, is to denature the krill enzymes and produce a denatured krill product, such as a krill meal, not to produce krill oil. *Id.* at 18, 20. Thus, we find that Budziński does not teach or suggest that krill should be denatured *and* their oil extracted immediately after hauling the krill on board a ship.

As noted in both Budziński and Breivik II, krill products are subject to both enzymatic degradation and oxidation. Ex. 1008, 6, 21–22; Ex. 1037, 2:6–9, 5:12–13, 6:7–9. Breivik II deals with these problems by, among other methods, denaturing krill at 80 °C and then immediately extracting krill oil from the denatured krill using a polar solvent and supercritical CO₂. Ex. 1037, 9:5–11. In contrast, Budziński cooks krill at 80–85 °C and then dries the product to form a stable krill meal. Ex. 1008, 20–21. For the reasons discussed above, we find that one of ordinary skill in the art would have understood that krill could be successfully extracted from a denatured krill product either on a ship or on land and that there are benefits to both land-based and ship-based extractions. Thus, we find that one of ordinary

skill in the art would have found it obvious to use either method for krill oil extraction.¹³

In view of the foregoing, and upon review of the record as a whole, we find that one of ordinary skill in the art would have sought to combine the disclosures of Catchpole, Budziński, Fricke, and Randolph to arrive at the subject matter of claim 1 of the '169 patent.

h) Conclusion with Respect to the Combination of Catchpole, Budziński, Fricke, and Randolph

Petitioner persuasively identifies where Catchpole, Budziński, Fricke, and Randolph, in combination, teach or suggest every limitation of claim 1 of the '169 patent. Petitioner also persuasively explains why one of ordinary skill in the art would have sought to combine the disclosures of Catchpole, Budziński, Fricke, and Randolph to arrive at the subject matter of claim 1 with a reasonable expectation of success. Accordingly, Petitioner demonstrates by a preponderance of the evidence that claim 1 would have been obvious over Catchpole, Budziński, Fricke, and Randolph.

6. Dependent Claims 4 and 14

Claims 4 and 14 depend from independent claims 1 and 12, respectively, and further require “wherein said denatured krill product is a

¹³ Patent Owner also reiterates its assertion that Budziński’s disclosure of a “stable” krill meal is not in the context of production of krill oil, “but use in animal feeds.” PO Resp. 46 (citing Ex. 1008, 31; Ex. 2015 ¶ 57). For the reasons discussed above, however, we find that one of ordinary skill in the art would have understood from the express disclosures of Budziński that its krill meal was stable and suitable for use as an animal feed or for krill oil extraction. Ex. 1008, 21–22 (noting that presence of astaxanthin in the krill meal and that krill meals were shown to be stable for 13 months of storage); *see also* Ex. 1032, 30 (Grantham disclosing that antioxidants can be used to successfully deal with lipid oxidation in krill products, should it be a factor).

krill meal.” Ex. 1001, 36:12–13, 36:43–44. Petitioner contends the subject matter of these claims would have been obvious in view of Budziński’s disclosure of “cooking and pressing or centrifuging and drying” krill to provide a processed, denatured krill meal with reduced moisture content that may be stored for up to 13 months. Pet. 54–55 (citing Ex. 1008, 24–26). Petitioner further contends that Fricke produces a denatured krill meal. *Id.* (citing Ex. 1010, 822–823).

Patent Owner asserts that Petitioner’s arguments do not establish obviousness of the challenged claims because Budziński does not teach extraction from krill meal after a storage period of from 1 to 24 months, and Fricke does not teach anything more than cooking whole krill, which Patent Owner contends does not demonstrate that a krill meal was produced, even under Petitioner’s broad construction of the term. PO Resp. 49 (citing Ex. 2015 ¶ 67).

Even if no prior art reference cited by Petitioner discloses within its four corners denaturing a krill product to produce a denatured krill meal, storing this product for between 1 and 24 months, and then extracting oil from this denatured krill meal using a polar solvent, the primary question under an obviousness ground is whether the prior art *as a whole* teaches or suggests the subject matter of the claim. On this record, Petitioner persuasively demonstrates that denaturing krill to form a krill meal was well known in the art and understood to result in a stable, denatured krill meal that can be stored for up to 13 months. Ex. 1008, 21–22 (Budziński). Petitioner also demonstrates that extraction of krill oil from krill products was known in the art (Budzinski and Catchpole), including from denatured krill products (Fricke). Moreover, Dr. Tallon persuasively testifies that one of ordinary skill in the art would have understood that extracting krill oil

from a freshly denatured krill product, or from a stable denatured krill meal that had been stored for several months, would be accomplished using the same process and it was understood that both methods would result in a useable krill oil. Ex. 1006 ¶ 171.

Accordingly, we find that one of ordinary skill in the art would have found it obvious to extract krill oil from a denatured krill meal product that had been stored for 1 to 24 months using a polar solvent. Thus, Petitioner demonstrates by a preponderance of the evidence that claims 4 and 14 would have been obvious over Budziński, Frick, Catchpole, and Randolph.

7. Claims 2, 3, 5–13, and 15–20

Dependent claims 2, 3, and 5–11 depend, directly or indirectly, from independent claim 1 and dependent claims 13 and 15–20 depend, directly or indirectly, from independent claim 12. Ex. 1001, 35:56–36:29. Petitioner contends the subject matter of claims 2, 3, 5–13, and 15–20 would also have been obvious over the combined disclosures of Budziński, Fricke, Catchpole, and Randolph. Pet. 49–59. In support of this contention, Petitioner provides a detailed explanation as to where each limitation of these claims is taught or suggested in one or more of Budziński, Fricke, Catchpole, and Randolph. *Id.*

Patent Owner does not contest Petitioner’s arguments with respect to claims 2, 3, 5–13, and 15–20, beyond its arguments set forth above with respect to independent claim 1.¹⁴ See PO Resp. 36–52.

¹⁴ Independent claim 12 is substantially identical to independent claim 1, differing only in that it requires denaturing “freshly harvested krill” as opposed to “krill.” Compare Ex. 1001, 35:45–47 with *id.* at 36:31–32. Both Budziński and Fricke disclose denaturing fresh caught krill, and Patent Owner does not argue otherwise. Ex. 1008, 20, 25; Ex. 1010, 822–823.

Upon review of the parties' arguments and supporting evidence, we find that Petitioner persuasively identifies where every limitation of claims 2, 3, 5–13, and 15–20 is taught or suggested in one or more of Budziński, Fricke, Catchpole, and Randolph. Accordingly, Petitioner demonstrates by a preponderance of the evidence that claims 2, 3, 5–13, and 15–20 would have been obvious over Budziński, Fricke, Catchpole, and Randolph.

G. Claims 6 and 16 over Catchpole, Budziński, Fricke, Randolph, and Sampalis I

Claims 6 and 16 depend from independent claims 1 and 12, respectively, and further require “encapsulating said krill oil.” Ex. 1001, 36:17–18, 36:47–48. Petitioner contends the subject matter of claims 6 and 16 would have been obvious over the combined disclosures of Catchpole, Budziński, Fricke, Randolph, and Sampalis I. Pet. 66–68.

1. Sampalis I

Sampalis I discloses the health benefits of an encapsulated krill oil product and, in particular, reports that two soft gels of the product taken daily “can significantly reduce the physical and emotional symptoms related to premenstrual syndrome.” Ex. 1012, Abst.

2. Analysis

Petitioner contends that one of ordinary skill in the art would have found it obvious to encapsulate krill oil extracts in view of Sampalis I's disclosure of encapsulating krill oil to form a medically beneficial krill oil product, and in view of the established understanding in the art that encapsulated krill oil extracts provide a “convenient and easy to administer dosage form.” Pet. 67–68 (citing Ex. 1006 ¶¶ 64, 69, 218).

Patent Owner argues that the ground based on Catchpole, Budziński, Fricke, Randolph, and Sampalis I fails because (1) Breivik II is not prior art

to the '169 patent, and (2) Sampalis I does not teach or suggest extracting krill oil from a denatured krill product that has been stored for from 1 to 24 months. PO Resp. 52–53.

Upon review of the parties' arguments and supporting evidence, we find that Sampalis I teaches or suggests encapsulating a krill oil to provide a "convenient and easy to administer dosage form." Patent Owner's argument based on Brevik II is not persuasive for the reasons discussed above. And, Patent Owner's counter-argument regarding Sampalis I's lack of disclosure is not on point because, as we discuss above, the combination of Catchpole, Budziński, Fricke, and Randolph teaches or suggests extracting krill oil from a denatured krill product that has been stored for from 1 to 24 months. Accordingly, Petitioner demonstrates by a preponderance of the evidence that claims 6 and 16 would have been obvious over Catchpole, Budziński, Fricke, Randolph, and Sampalis I.

III. REPLY ARGUMENTS

Patent Owner sought authorization to file a motion to strike portions of Petitioner's Reply. We denied this request, but permitted Patent Owner to file a three-page paper identifying any evidence or argument in the Reply that it considered to be improper. Paper 19, 2. We also authorized Petitioner to file a three-page response to Patent Owner's submission. *Id.*

A. *"Some Months" Argument and Prior Board Decision*

The Petition asserts that Fricke teaches or suggests storage of processed krill prior to solvent extraction. Pet. 45 (citing Ex. 1010, 2–3; Ex. 1006 ¶ 444). In response, Patent Owner argued that Fricke "contains no teaching as to the storage time of the cooked krill." PO Resp. 38. Petitioner replied to this argument by noting that in IPR2017-00746 the Board previously found that the samples of Fricke were extracted after being stored

for “some months.” Pet. Reply 20–21 (citing Ex. 1104, 28–29; Ex. 2006; Ex. 1086 ¶¶ 62–65).

Patent Owner contends the argument that Fricke’s denatured krill samples were stored for “some months,” as well as Petitioner’s citation to Exhibit 1160, which was first filed with Petitioner’s Reply, constitute improper new arguments and evidence “intended to supplement Petitioner’s arguments” in the Petition. Paper 19, 2. Patent Owner further contends that the Board’s decision in IPR2017-00746 was not cited in the Petition and is improperly intended to supplement Petitioner’s arguments in the Petition. *Id.* at 2–3.

For the reason set forth in footnote 12 above, we need not rely on Petitioner’s citation to Exhibit 1160 or the Board’s previous finding regarding the storage time for the krill of Fricke in finding the challenged claims unpatentable. As such, Patent Owner’s objections are moot.

B. Grounds Without Breivik II

In its Reply, Petitioner asserts that one or more alternative references are provided for every claim limitation for which Breivik II is cited. Pet. Reply 7. Patent Owner contends “this statement implies that there are Grounds that do not include Breivik II,” which “is the first reference cited in both instituted Grounds.” Paper 19, 3.

As noted by Patent Owner, each ground asserted by Petitioner includes Breivik II. Pet. 10. In each ground, however, Petitioner relies upon multiple references as teaching or suggesting the subject matter recited in the challenged claim. *See id.* at 44. A petitioner may, as Petitioner has done here, rely on multiple references each of which independently establishes that a particular claim limitation was known in the art. As such, Petitioner’s explanation that multiple references are relied upon for each claim limitation

is not an improper reply argument, but instead an accurate characterization of how the grounds are presented in the Petition.

IV. PETITIONER’S MOTION TO EXCLUDE

Petitioner moves to exclude Exhibits 2003, 2010, and 2013, as well as paragraphs 7, 8, 12–14, and 17 of the Declaration of Dr. Tilseth (Ex. 2001) and paragraph 28 of the Declaration of Dr. Jaczynski (Exhibit 2015). Paper 25, 1.

Each of the documents and paragraphs identified by Petitioner relates to Patent Owner’s attempt to swear behind Breivik II to remove it as a prior art reference in this proceeding. *Id.* at 2–13. As we do not address this issue in this Decision, Petitioner’s Motion to Exclude is dismissed as moot.

V. CONCLUSION¹⁵

In consideration of the above, we find that Petitioner demonstrates by a preponderance of the evidence that claims 1–5, 7–15, and 17–20 are unpatentable as obvious over the combined teachings of Breivik II, Catchpole, Budziński, Fricke, and Randolph, and that claims 6 and 16 are unpatentable as obvious over the combined teachings of Breivik II, Catchpole, Budziński, Fricke, Randolph, and Sampalis I.

¹⁵ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis¹⁶	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–5, 7–15, 17–20	103(a)	Breivik II, Catchpole, Budziński, Fricke, Randolph	1–5, 7–15, 17–20	
6, 16	103(a)	Breivik II, Catchpole, Budziński, Fricke, Randolph, Sampalis I	6, 16	
Overall Outcome			1–20	

VI. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–20 of the '169 patent are unpatentable;

FURTHER ORDERED that Petitioner's Motion to Exclude is dismissed as moot; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

¹⁶ As noted above, we have considered Petitioner's grounds without relying on Breivik II. We do, however, consider whether Breivik II would teach away from or counsel against the combinations of prior art asserted in the Petition. *See* Part II.F.5.g *supra*.

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